

Analysis of Feedback Mechanisms and k-factors in ADSRs

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Physics of Subcritical Multiplication

Fundamental Kinetic Parameters

Multiplication Factors

- k_{eff} (**Eigenvalue**): Ratio between production rate and total loss rate. It is the eigenvalue of the fundamental mode. Determines system criticality in the absence of a source.
- k_s (**Source Factor**): Real multiplication "seen" by the source (N_{fiss}/N_{tot}). Spatially dependent on the source position and its energy.

Importance and Gain

- **Importance** ($\phi^\dagger$): Effectiveness of the source neutron in contributing to the total population:

$$\phi^\dagger(r, E) = \frac{k_s(r, E)}{1 - k_s(r, E)}$$

- **Gain** (G): Beam energy amplification:

$$G = \frac{G_0 k}{1 - k}$$

Relationship between k_{eff} and k_s

The Role of Source Efficiency φ^*

In a subcritical system, k_{eff} and k_s do not coincide. Their relationship is mediated by the source efficiency φ^* , which quantifies the importance between external and fission neutron.

Equivalence Equation

$$\left(1 - \frac{1}{k_{eff}}\right) = \varphi^* \left(1 - \frac{1}{k_s}\right)$$

Physical Meaning of φ^* :

- $\varphi^* > 1$: The source is located in a high-importance zone (core center), minimizing leakage.
- $\varphi^* < 1$: The source is external or in a peripheral zone (low efficiency).

Operational Implications:

- Optimizing target position to achieve $\varphi^* > 1$ reduces the necessary beam current to maintain the same thermal power (P_{th}).

Fuel Evolution

Long-term Feedback

Isotopic Dynamics (Bateman Eq.)

The evolution of n_i is governed by the balance between production and destruction:

$$\frac{dn_i(t)}{dt} = \sum_{j \neq i} (\sigma_{j \rightarrow i} \phi + \lambda_{j \rightarrow i}) n_j - (\sigma_{i, tot} \phi + \lambda_{i, tot}) n_i$$

Spectral Comparison on $k(t)$ stability:

Fast Spectrum

Breeding ($^{232}\text{Th} \rightarrow ^{233}\text{U}$) almost exactly compensates for poisons. **Result:** k extremely stable for long cycles.

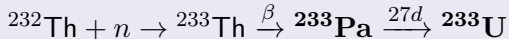
Thermal Spectrum

Poison accumulation dominates over breeding. **Result:** Intrinsic instability; k drops drastically in < 1 year.

Reactivity and Passive Safety

Short-term Effects

The Protactinium Effect (^{233}Pa) in the Th - U cycle



- **During operation:** ^{233}Pa acts as a strong neutron poison ($\sim 50b$).
- **At shutdown:** ^{233}Pa decays into fissile ^{233}U , causing a positive reactivity insertion $\Delta k > 0$.
- **Constraint:** Limits maximum operational flux and nominal k to avoid accidental criticality post-shutdown.

Xenon Effect (^{135}Xe)

Dominates in thermal reactors (post-shutdown peak), but is **negligible** in fast spectrum.

$$(\sigma_c(\text{Th}) = 2.36 \times 10^6 \text{ b})$$

Temp. ($\alpha_T = \Delta k / \Delta T$)

Instantaneous feedback. Must be negative ($\Delta T \uparrow \Rightarrow k \downarrow$) to ensure intrinsic safety.

Regulations of ADSRs

Regulatory Statement (D.Lgs. 101/2020)

There is no legal limit for the k_{eff} value, but ISIN/IAEA technical practice applies.

- **The Issue is Uncertainty (σ):** Designing at $k_{eff} = 0.999$ is legally unsustainable. With a typical error of ± 0.005 , the system is indistinguishable from a critical one ($k \geq 1$).
- **Operational Limits for the "Subcritical" License:**
 - **Storage/Transport:** $k_{eff} \leq 0.95$ (5% Margin).
 - **Operational (ADS/Research):** $k_{eff} \leq 0.97 - 0.98$ (3σ Rule).

The Strategy I Would Follow... (k_{eff} vs k_s)

The law regulates intrinsic safety (k_{eff}), not the gain with source (k_s). **Goal:** Maintain $k_{eff} \approx 0.975$ (legal) and maximize source importance (φ^*) to achieve high energy gain.

Bibliographic References

The information presented was mainly elaborated starting from the following:

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④ **Nifenecker, H., David, S., Loiseaux, J.M., Meplan, O.** (2001). *Basics of accelerator driven subcritical reactors*. Nuclear Instruments and Methods in Physics Research A, 463(3), 428-467.